SharePoint Online & OneDrive → Google Drive / Shared Drives Migration

Implementation Solution Guide

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Executive Summary — SharePoint Online & OneDrive → Google Drive / Shared Drives Migration

Moving files off SharePoint Online and OneDrive to Google Drive is not a copy job — it is a **mapping job**, and permissions and sharing are where it goes wrong. A SharePoint site does not map one-to-one onto Google Drive: team content dumped into a personal My Drive gets one owner who can lock everyone out; permission levels flatten so a team that should still edit lands as view-only; sharing links break or, worse, widen; and any identity that was not mapped has its access **silently dropped**, so the file arrives with no permissions at all — none of it discovered until a user reports they cannot open their own work. This solution removes those failure modes.

This is a complete, self-guided package written for a competent migration admin. It supplies the **mapping model** (which sites/libraries become which Shared Drives, whose OneDrive becomes whose My Drive), the **permission and sharing translation** (SharePoint levels → Google roles, SharePoint sharing → Google sharing, via a complete identity map), the **tooling pattern** (tool-agnostic — the engine is chosen in the design and its internals deferred to the vendor), and the **per-item validation** that proves each file landed in the right place with the right access. It is executed **pilot-first** and **rollback-ready**.

What you get

• A **Solution Overview** of the SPO/OneDrive → Drive/Shared Drives mapping model — sites/libraries →

Shared Drives, OneDrive → My Drive, the permission level → role translation, and the sharing translation rules.

• A **Requirements Guide** with verify / implement-if-missing for every prerequisite: source

SPO/OneDrive access, a Google tenant with Shared Drives and storage, a chosen migration tool, a complete identity map, the mapping decisions, and a named pilot.

• An **Implementation Guide** of precise, ordered steps: discover the source, define the content /

permission / identity / sharing mapping, prove it on a named pilot with per-item validation, fix the mapping (not just the item), then migrate wave by wave and validate each wave before the next.

• **Five PowerShell scripts** — source content/permission discovery, per-item migration + permission

validation, a coverage/exception report, safe remediation, and a wave target-group builder — all read-only against Microsoft Graph except the two write scripts, which default to `-WhatIf`.

• **Testing and Troubleshooting** guides with explicit per-item gates (files migrated, permissions

correct, sharing works), plus Best Practices for shared-drive structure — and a migration playbook outline the migration runs from.

The solution keeps the migration engine **tool-agnostic**: it owns the mapping, translation, and validation, so a change of tool touches the engine, not the method.

What this delivers

A governed, validated Drive migration: SharePoint Online and OneDrive content on Google Drive / Shared Drives with **permissions and sharing intact**, mapped deliberately and validated per item, proven on a pilot before any business unit's content moves.

The deployment model

Assess, then Configure, then Pilot, then Validate, then Expand — with per-item validation of files, permissions, and sharing before every wave passes, and source content set read-only only after validation is clean.

Outcome

Files and shares land on Google Drive / Shared Drives with the **right access** — team content in Shared Drives, personal content in My Drive, roles preserved, sharing translated, and nothing silently dropped or widened — proven on a pilot and validated wave by wave.

1. Solution Overview — SharePoint Online & OneDrive → Google Drive / Shared Drives Migration

What this solution is

This solution migrates **SharePoint Online (SPO) and OneDrive for Business** content to **Google Drive and Shared Drives** — with the content, folder structure, **permissions, and sharing** mapped deliberately rather than dumped file-for-file. The hard part of a Drive migration is never the bytes; it is that a SharePoint site and a OneDrive do **not** map one-to-one onto Google Drive, and that permissions and sharing are where these migrations quietly go wrong. This solution supplies the **mapping model** (what becomes a Shared Drive, what becomes a folder, whose OneDrive becomes whose My Drive), the **permission/sharing translation** (SharePoint permission levels → Google roles, SharePoint sharing → Google sharing), the **tooling pattern** (tool-agnostic — the migration engine is chosen and its internals deferred to the vendor), and the **per-item validation** that proves each item landed in the right place with the right access. It is executed **pilot-first** and **rollback-ready**.

Why this matters

Drive migrations fail on access, not volume. A team's SharePoint site is copied into a personal My Drive and now has one owner who can lock everyone out. Permission levels are flattened — a SharePoint "Contribute" that should have become a Google **Contributor** lands as **Viewer**, and the team can no longer edit. Sharing links break: an "anyone with the link" SharePoint share does not carry across, or worse, carries across too broadly. Identities are not mapped, so permissions granted to `jsmith@source` never resolve on the Google side and are silently dropped — the file arrives with no access at all ([CloudM — Drive permission changes](https://support.cloudm.io/hc/en-us/articles/25031144447644-Google-Drive-Permission-Changes-Microsoft-365-to-Google-Workspace)). And because nothing was validated per item, none of this is discovered until a user reports they cannot open their own files — weeks after the tool declared success. This solution fixes the cause: it decides the content and permission mapping **up front**, proves it on a **named pilot**, validates files, permissions, and sharing **per item**, and only then migrates the rest **by wave** — each wave validated before the next.

The SPO/OneDrive → Drive/Shared Drives mapping model

The migration is driven by an explicit mapping. Author it once, name it, and reuse it for every wave.

Source (Microsoft 365)	Target (Google)	Rationale
SharePoint team site / document library (team-owned content)	**Shared Drive**	Team content must be team-owned. A Shared Drive has no single personal owner, so access survives when a person leaves — the correct home for site/library content.
SharePoint site with several libraries	**One Shared Drive per library**, or one Shared Drive with a folder per library	Keeps Shared Drives within item/member limits and gives each library a clean access boundary.
OneDrive for Business (personal content)	The same user's **My Drive**	Personal content stays personal, owned by the same identity.
SharePoint folder	**Folder inside the target Shared Drive**	Structure is preserved beneath the Shared Drive root.
SharePoint permission level	**Google Drive role** (see translation below)	Access intent is preserved, not flattened.
SharePoint / OneDrive share (direct grant or link)	**Google sharing** (member/permission or link)	Sharing intent is translated deliberately, not lost or over-broadened.

The named mapping artifacts:

Artifact	Name
Content + structure mapping	`GWDRIVE Mapping - Content & Structure - Source to Target`
Permission/role translation table	`GWDRIVE Mapping - Permissions - Level to Role`
Identity mapping table	`GWDRIVE Mapping - Identity - Source to Google`
Sharing translation rules	`GWDRIVE Mapping - Sharing - Link & Grant Rules`
Pilot scope	`GWDRIVE Pilot - Scope - Wave 0`
Per-wave migration + validation runbook	`GWDRIVE Runbook - Migrate & Validate - Per Wave`

Permission translation — SharePoint level → Google role

Google Shared Drives have five member access levels; the migration maps each source permission level to the closest Google role so **access intent is preserved** rather than flattened ([Google — shared drive

roles](https://developers.google.com/workspace/drive/api/guides/ref-roles)).

SharePoint permission level (typical)	Google Drive role	What the role grants
Full Control / site owner	**Manager** (`organizer`)	Manage files, folders, members, and settings; delete; move out of the drive
Design / Edit (manage content)	**Content manager** (`fileOrganizer`)	Add, edit, move, and organize content; share items; cannot manage members
Contribute	**Contributor** (`writer`)	View, comment, edit, and create files; cannot manage members or settings
Read + comment	**Commenter**	View and comment only
Read	**Viewer** (`reader`)	View only

Exact source permission-level names vary per SharePoint tenant; the translation table is authored against the source tenant's **actual** levels, not assumed defaults, and reviewed with the content owners before the pilot.

Sharing translation

SharePoint sharing (direct grants and sharing links) is translated by rule into Google sharing:

• **Direct grants** to named users/groups → Google **permissions** on the item or membership on the

Shared Drive, via the **identity mapping table** so each source principal resolves to a Google identity. **Any principal missing from the identity map is dropped** by the tool and its permission is lost — so the identity map must be complete before migrating ([CloudM — Drive permission changes](https://support.cloudm.io/hc/en-us/articles/25031144447644-Google-Drive-Permission-Changes-Microsoft-365-to-Google-Workspace)).

• **Organization sharing links** ("people in your organization") → Google **domain** sharing at the

matching role.

• **Anyone-with-the-link** shares → translated **only where policy allows**; otherwise down-scoped to

domain or named access and flagged for owner review, so a migration does not silently widen access.

• **Note on visibility:** as of early 2026, folders a user could not see in SharePoint may appear

greyed out (visible but inaccessible) in Google Drive — visibility changed, not access — so users are told to expect this and restricted content is best placed in separate top-level folders before migrating.

The deployment model

> *Define the content, permission, identity, and sharing mapping and confirm access; prove the whole mapping on a **named pilot**; validate files, permissions, and sharing **per item**; then migrate **by wave** and validate each wave before the next — the repeatable way: Assess, then Configure, then Pilot, then Validate, then Expand.*

What this is not

This solution governs the **mapping, translation, and per-item validation** of an SPO/OneDrive → Drive migration. It does **not** re-document the internal console steps of any specific migration engine — the tool is chosen in the design and its mechanics are deferred to the vendor's own docs (tool-agnostic). Where the included scripts read the source, they read **SharePoint Online and OneDrive through Microsoft Graph, read-only**; they move no content and change no permissions (the one write script and the scope-targeting script default to `-WhatIf`).

2. Requirements Guide — SharePoint Online & OneDrive → Google Drive / Shared Drives Migration

Verify every requirement before starting the migration. Each entry gives **Description**, **Why**, **How to verify**, and **How to implement if missing**. Category-wide prerequisites (the Google Workspace tenant, the source Microsoft 365 / Entra environment, DNS control, the identity source of truth, a pilot scope, and governance stakeholders) are in `global-requirements-google.md`; this guide adds what a Drive migration specifically needs.

2.1 Source SharePoint Online & OneDrive access

• **Description:** Administrative read access to the source **SharePoint Online** sites/libraries and **OneDrive for Business** in scope — through the Microsoft 365 admin roles and the Microsoft Graph read scopes the included scripts request (`Sites.Read.All`, `Files.Read.All`, `User.Read.All`, `Group.Read.All`).

• **Why required:** Discovery reads the source content, structure, and permissions to build the mapping; the migration tool reads the source to move it. Without this, the migration cannot see what it is moving.

• **How to verify:** Sign in with the migration admin account and confirm the in-scope sites and OneDrives are enumerable; run `GWDRIVE-Discovery.ps1`.

• **How to implement if missing:** Obtain the SharePoint/Global admin roles (or a scoped migration account) from the source tenant owner and consent to the read scopes on first script run.

2.2 Target Google Workspace tenant & licensing

• **Description:** A provisioned Google Workspace tenant (per the global requirements) with **Shared Drives enabled** for the target org units and enough Drive storage/licensing for the content being moved.

• **Why required:** Team content lands in Shared Drives and personal content in each user's My Drive; both must exist and have storage headroom before content arrives.

• **How to verify:** In the Google Admin console, confirm Shared Drive creation is enabled for the target OUs and that pooled/assigned storage covers the discovered source volume.

• **How to implement if missing:** Enable Shared Drives for the target OUs and provision storage / licensing (through Google or an authorized reseller) before the pilot.

2.3 A migration tool (chosen, tool-agnostic)

• **Description:** A selected third-party Drive migration engine capable of moving SPO/OneDrive content to Google Drive/Shared Drives **with permission and sharing mapping and identity mapping**, authorized against both tenants.

• **Why required:** This solution supplies the mapping, translation, and validation; the engine does the byte movement. Choosing it up front (not per wave) keeps outcomes consistent.

• **How to verify:** The tool is selected, authorized to read the source and write the target, and its identity/permission-mapping capability is confirmed in a test move.

• **How to implement if missing:** Select and authorize the engine, connect both tenants, and prove a single test item moves with permissions in a lab before the pilot. The engine's internal console steps are deferred to the vendor's own documentation.

2.4 Identity mapping — source principals → Google identities

• **Description:** A complete **identity mapping table** — `GWDRIVE Mapping - Identity - Source to Google` — mapping every source domain, group, and user that holds a permission or share to its Google counterpart.

• **Why required:** Permissions and shares are re-applied to the mapped Google identity; **any source principal missing from the map has its permission silently dropped**, so the item arrives with lost access. Completeness of this table is the single largest determinant of a clean permission result ([CloudM — Drive permission changes](https://support.cloudm.io/hc/en-us/articles/25031144447644-Google-Drive-Permission-Changes-Microsoft-365-to-Google-Workspace)).

• **How to verify:** Every principal discovered on in-scope content resolves to a Google identity; run `GWDRIVE-Discovery.ps1` to enumerate principals and `GWDRIVE-Validation.ps1` to confirm coverage.

• **How to implement if missing:** Build the map from the discovery output (or your identity/provisioning source of truth) and fill every gap before migrating; use `GWDRIVE-Remediation.ps1` (`-WhatIf` by default) to help stage additions.

2.5 Mapping decisions — content, permissions, sharing

• **Description:** The authored, owner-approved mapping set: the **content/structure** map (`GWDRIVE Mapping - Content & Structure - Source to Target` — which sites/libraries become which Shared Drives, which OneDrives become which My Drives), the **permission/role translation table** (`GWDRIVE Mapping - Permissions - Level to Role`), and the **sharing translation rules** (`GWDRIVE Mapping - Sharing - Link & Grant Rules`).

• **Why required:** SPO/OneDrive does not map one-to-one onto Drive; deciding the mapping per wave (or

letting the tool default it) produces inconsistent structure and flattened permissions. The mapping is the design every wave executes.

• **How to verify:** Each in-scope site/library/OneDrive has a named target; each source permission

level has a Google role; each sharing type has a rule — all reviewed with the content owners.

• **How to implement if missing:** Author the three mapping artifacts in Implementation Step 2 against

the source tenant's **actual** permission levels and sharing settings, and freeze them before the pilot.

2.6 A named pilot and change control

• **Description:** A named pilot — `GWDRIVE Pilot - Scope - Wave 0` — of a representative SharePoint

site (or library) and a small set of OneDrives, with a change-control window for the production target changes.

• **Why required:** The pilot proves the whole mapping (content, permissions, identity, sharing) end

to end at low blast radius before any business unit's content moves; change control protects the production Google tenant.

• **How to verify:** A representative pilot scope is named and agreed, and a change window/approval

path exists (aligns to the global-requirements pilot and change-control items).

• **How to implement if missing:** Choose a representative site + OneDrive sample as Wave 0 and agree

the change-control process with the content and platform owners before migrating.

Checklist

- Source SPO/OneDrive read access confirmed — `GWDRIVE-Discovery.ps1`
- Target Google tenant: Shared Drives enabled + storage/licensing sufficient
- Migration tool selected, authorized both tenants, permission-mapping proven in a test move
- Identity map complete — every source principal resolves to Google — `GWDRIVE-Validation.ps1`
- Content/structure map authored — `GWDRIVE Mapping - Content & Structure - Source to Target`
- Permission/role translation table authored — `GWDRIVE Mapping - Permissions - Level to Role`
- Sharing translation rules authored — `GWDRIVE Mapping - Sharing - Link & Grant Rules`
- Named pilot + change control agreed — `GWDRIVE Pilot - Scope - Wave 0`

Global Requirements — Google Workspace & Gemini Solutions

These requirements apply to **every** solution in the **Google** category. They are the common foundation each Google Workspace / Gemini solution assumes is already in place (or stands up as part of its own Requirements section). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every Google solution).*** *The **Google Admin console** (admin.google.com), **Google Cloud / Cloud Identity**, and the **Microsoft / Entra** sides are authored at exact, current click-by-click detail from the vendors' public documentation. **Gemini Enterprise** capabilities that are in preview or rolling out are **cited or flagged "verify in your tenant"** — never invented. **Third-party migration tools** (e.g. for mail/Drive/Chat data movement) and **third-party enterprise apps** (ServiceNow, SAP, Databricks, etc.) are integrated at the pattern/architectural level; their internal console steps are deferred to the vendor's own docs.*

1. Google Workspace / Cloud Identity tenant

• **Description:** A provisioned Google Workspace (or Cloud Identity) tenant with a **super-admin** account for configuration, and the appropriate Workspace edition/licensing for the capability in scope (e.g. Enterprise Plus for Vault/CAA/DLP, Gemini/AI add-on where required).

• **Why required:** All configuration, policy, and API work is performed against this tenant.

• **How to verify:** Sign in to **admin.google.com** as a super-admin; confirm edition and license counts under **Billing / Subscriptions**.

• **How to implement if missing:** Provision the tenant and licenses through Google or an authorized reseller before starting.

2. Source Microsoft 365 / Entra environment (for migration & coexistence)

• **Description:** For migration/coexistence/identity solutions, administrative access to the source **Microsoft 365** tenant and **Microsoft Entra ID** (Global Administrator or the least-privilege roles the solution names).

• **Why required:** Migration, directory sync, SSO federation, and coexistence read from and change the Microsoft side.

• **How to verify:** Confirm access to the M365 admin center and the Entra admin center with the required role.

• **How to implement if missing:** Obtain the role from the M365/Entra tenant owner.

3. DNS control for the routed domains

• **Description:** Ability to create/modify **DNS records** (MX, SPF, DKIM, DMARC, TXT, CNAME) for the domains being migrated, coexisting, or verified.

• **Why required:** Domain verification, mail routing/coexistence, and authentication all depend on DNS changes.

• **How to verify:** Confirm access to the authoritative DNS zone for each domain.

• **How to implement if missing:** Coordinate with whoever controls DNS; stage changes under change control (mail-flow changes are high-impact).

4. Identity & provisioning source of truth

• **Description:** A defined **identity source of truth** (on-prem AD, Entra ID, or an HR/IGA system) and the sync mechanism (e.g. **Google Cloud Directory Sync (GCDS)** or SCIM-based provisioning).

• **Why required:** Consistent users/groups across ecosystems underpin SSO, licensing, and access.

• **How to verify:** Confirm which directory is authoritative and how objects flow into Google.

• **How to implement if missing:** Define the source of truth and provisioning model before configuring SSO or migrating.

5. A pilot scope (OU / group) and change control

• **Description:** A **pilot organizational unit or group** to validate on before fleet-wide rollout, and a **change-control** process for production changes (especially mail flow, DNS, identity, and DLP).

• **Why required:** Every solution follows an **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** model; the pilot proves the design and change control protects production.

• **How to verify:** Confirm a named pilot OU/group exists and a change window/approval path is available.

• **How to implement if missing:** Create a pilot OU/group and agree the change-control process with the owners before making production changes.

6. Security, compliance & governance stakeholders

• **Description:** Engagement with the **security (CISO), legal/eDiscovery, and data-governance** stakeholders whose policies the configuration must satisfy (DLP, retention, Vault/Purview, and any regulatory regime such as ITAR/DFARS/Assured Controls where applicable).

• **Why required:** Data governance, eDiscovery, DLP, and regulated-workload settings must align to enterprise policy and legal requirements, not just technical defaults.

• **How to verify:** Confirm the applicable policies and named stakeholders/approvers.

• **How to implement if missing:** Identify the owners and obtain the policy/compliance requirements before configuring governance controls.

*The deployment model across every Google solution is the same: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**, with explicit pass/fail gates and a documented rollback for production changes.*

3. Implementation Guide — SharePoint Online & OneDrive → Google Drive / Shared Drives Migration

Follow these steps in order. This guide defines the content and permission mapping, proves it on a **named pilot**, validates files, permissions, and sharing **per item**, then migrates **by wave** and validates each wave before the next. Artifact names are fixed — create them with the exact names shown and reuse them for every wave. The migration engine's internal steps are **tool-agnostic** and deferred to the vendor's own docs; this guide governs the mapping, translation, and validation.

> *Model: define the mapping and confirm access, prove it on a pilot, validate per item, then expand wave by wave. The repeatable way: Assess, then Configure, then Pilot, then Validate, then Expand.*

Step 1 — Assess: discover the source content, structure, and permissions

1. Run `GWDRIVE-Discovery.ps1` to inventory the in-scope SharePoint sites/libraries and OneDrives:

item and folder counts, structure, the **permission principals** on each scope, and the **sharing** in place (organization/anyone links). Record it as the discovery baseline.

1. Extract the **distinct set of principals** (users, groups, domains) that hold any permission or

share across the in-scope content — this is the input to the identity map.

1. Record gaps against the Requirements checklist. Do not proceed past missing source access or a

target tenant without Shared Drives enabled — resolve those first.

Step 2 — Configure: define the content, permission, identity, and sharing mapping

1. **Content + structure map** — author `GWDRIVE Mapping - Content & Structure - Source to Target`:

for each SharePoint site/library decide its target **Shared Drive** (one per library where a site has several, to stay within the 500,000-item and 600-member Shared Drive limits and keep clean access boundaries), and map each **OneDrive** to the **same user's My Drive**. Preserve folder structure beneath each target root.

1. **Permission/role translation table** — author `GWDRIVE Mapping - Permissions - Level to Role`

against the source tenant's **actual** permission levels: Full Control → **Manager**, manage-content levels → **Content manager**, Contribute → **Contributor**, read+comment → **Commenter**, read → **Viewer** ([Google — shared drive roles](https://developers.google.com/workspace/drive/api/guides/ref-roles)). Have the content owners confirm the mapping preserves their intended access.

1. **Identity map** — author `GWDRIVE Mapping - Identity - Source to Google` covering **every**

principal from Step 1's set. A principal missing here loses its permission on migration — treat the map's completeness as a gate, not a nicety.

1. **Sharing translation rules** — author `GWDRIVE Mapping - Sharing - Link & Grant Rules`: direct

grants → Google permissions/membership (via the identity map); organization links → domain sharing at the matching role; anyone-with-the-link → translated only where policy permits, else down-scoped and flagged for owner review. Freeze all four artifacts before the pilot.

Step 3 — Configure the target and the tool

1. Pre-create (or let the tool create) the target **Shared Drives** named per the content map, and

confirm each user's **My Drive** exists for OneDrive targets.

1. Load the **identity map**, **permission translation**, and **sharing rules** into the chosen

migration engine (per the vendor's own docs). Confirm the engine will apply folder-level permissions to folders and file-level permissions to files where the source differs — the accurate default — reserving root/drive-level grants only where the top-level access should cascade.

1. Prove one test item moves with its permission and sharing correct before running the pilot at

volume.

Step 4 — Pilot: migrate Wave 0 and validate per item

1. Confirm the pilot scope `GWDRIVE Pilot - Scope - Wave 0` (a representative site/library plus a small

OneDrive sample) and build its target group with `GWDRIVE-Set-ScopeTargeting.ps1` (runs with `-WhatIf` by default; re-run with `-WhatIf:$false` to apply).

1. Run the migration engine over the pilot scope under change control, applying the four mapping

artifacts.

1. Run `GWDRIVE-Validation.ps1` to validate the pilot **per item**: every source item has a target

item (files present), each item's permissions match the translated roles, and sharing resolved per the rules — with no dropped principals.

1. Run `GWDRIVE-Report.ps1` for pilot coverage and the exception list.

Step 5 — Validate the pilot and fix the mapping, not just the item

1. Treat every validation exception as a **mapping** defect: a flattened permission means the

translation table is wrong; a dropped grant means the identity map is incomplete; a mis-placed folder means the content map is wrong. Fix the **artifact**, so later waves inherit the fix.

1. Remediate identity-map / permission gaps with `GWDRIVE-Remediation.ps1` (runs with `-WhatIf` by

default; add `-WhatIf:$false` to act), then re-run `GWDRIVE-Validation.ps1` until the pilot is clean. The pilot passes only when files, permissions, and sharing all validate.

Step 6 — Build the wave plan

1. Segment the remaining in-scope content into sized waves — a coherent set of sites/libraries and

OneDrives per wave — so no single wave exceeds the tool's throughput or a Shared Drive's limits. Keep each wave a coherent business unit so comms and support are focused.

1. Build each wave's target group with `GWDRIVE-Set-ScopeTargeting.ps1` (preview by default). Assign

every source scope to exactly one wave; never move a confirmed scope between waves (risks duplicate content).

Step 7 — Expand: migrate by wave and validate each wave

1. For each wave, apply the same runbook `GWDRIVE Runbook - Migrate & Validate - Per Wave`: build the

wave group, run the engine with the four mapping artifacts, then run `GWDRIVE-Validation.ps1` (per-item files/permissions/sharing) and `GWDRIVE-Report.ps1` (coverage + exceptions).

1. A wave passes only when its per-item validation is clean. Remediate exceptions with

`GWDRIVE-Remediation.ps1` (`-WhatIf` by default) before passing the wave and starting the next.

1. Keep the identity map current as later waves introduce new principals; a stale map drops their

permissions.

Step 8 — Close out and cut over

1. When all waves validate, confirm users can access their migrated content at the mapped roles, brief

them on the greyed-out-folder visibility change, and set the source content to read-only (or decommission) per the change plan — never before validation is clean.

1. Retain the discovery baseline, mapping artifacts, and final `GWDRIVE-Report.ps1` coverage as the

record of what moved and where.

Order of operations

```
Assess/discover -> Define mapping (content+permission+identity+sharing) ->
Configure target+tool -> Pilot (Wave 0) + per-item validate -> Fix the mapping
-> Wave plan -> Expand wave by wave + validate each -> Close out / cut over
```

5. Testing Guide — SharePoint Online & OneDrive → Google Drive / Shared Drives Migration

Run these validation gates for **every** scope — first for Wave 0 (the pilot), then for each expansion wave — before that wave is declared done. Each test confirms one part of the mapping (files, permissions, sharing, structure, identity) worked for the wave's content. A wave does not pass until its gate criteria are met. Use `GWDRIVE-Validation.ps1` for the per-item checks and `GWDRIVE-Report.ps1` for coverage and exceptions.

Test 1 — Discovery and mapping are complete

Do: Confirm every in-scope site/library/OneDrive in the wave has a named target in `GWDRIVE Mapping - Content & Structure - Source to Target`, every source permission level has a role in `GWDRIVE Mapping - Permissions - Level to Role`, and every discovered principal resolves in `GWDRIVE Mapping - Identity - Source to Google`; run `GWDRIVE-Discovery.ps1`. Expected: No in-scope scope, permission level, or principal is unmapped for the wave.

Test 2 — Files migrated (content coverage)

Do: For the wave's scopes, compare source item/folder counts against the target; run `GWDRIVE-Validation.ps1`. Expected: Every source file and folder has a corresponding target item in the mapped Shared Drive / My Drive; item counts reconcile within the tool's known exclusions, and no scope is empty.

Test 3 — Permissions correct (roles preserved, not flattened)

Do: For a sample of items across each permission level, confirm the target Google role matches the translated level in `GWDRIVE Mapping - Permissions - Level to Role`; run `GWDRIVE-Validation.ps1`. Expected: Full Control → Manager, manage-content → Content manager, Contribute → Contributor, read+comment → Commenter, read → Viewer — access intent preserved, nothing flattened to Viewer.

Test 4 — No dropped principals

Do: For the sampled items, confirm every source grantee has a corresponding Google grant (no principal was silently dropped for want of an identity mapping). Expected: Every permission on the source item is present on the target item at the mapped role; any dropped grant traces to a gap in the identity map, which is then closed.

Test 5 — Sharing translated correctly

Do: Check the wave's shared items: direct grants, organization links, and anyone-with-the-link shares against `GWDRIVE Mapping - Sharing - Link & Grant Rules`. Expected: Direct grants map to Google permissions/membership; organization links map to domain sharing at the matching role; anyone-with-the-link shares are translated only where policy allows and otherwise down-scoped and flagged — no share is silently widened or lost.

Test 6 — Structure preserved

Do: Spot-check folder hierarchy for a sample of migrated scopes. Expected: Folder structure beneath each target root matches the source; nesting is within the Shared Drive limit and no folders collapsed or re-parented incorrectly.

Test 7 — Users can access their content

Do: Have a sample of wave users open their migrated content (team content in the Shared Drive, personal content in My Drive) at their expected role. Expected: Users can view/edit/manage per their mapped role;

owners can manage Shared Drive members; users are briefed that previously-hidden folders may appear greyed out (visibility change, not access).

Test 8 — Coverage and exceptions reported

Do: Run `GWDRIVE-Report.ps1` for the wave. Expected: The report shows content coverage (items migrated vs source), permission/sharing validation pass rate, and an itemized exception list with no unresolved exceptions before the wave passes.

Pass/fail gate

Gate	Requirement
Wave 0 (pilot) pass	Tests 1-8 pass; every exception is fixed in the **mapping artifact** (not just the item) before expanding
Expansion wave pass	Tests 1-8 pass for the wave; the exception list from `GWDRIVE-Report.ps1` is empty before the next wave starts
Migration complete	All waves passed, users confirmed access at mapped roles, and source content set read-only / decommissioned only after validation was clean

6. Troubleshooting Guide — SharePoint Online & OneDrive → Google Drive / Shared Drives Migration

Use `GWDRIVE-Discovery.ps1` for the source content/permission inventory, `GWDRIVE-Validation.ps1` for per-item files/permissions/sharing validation, `GWDRIVE-Report.ps1` for coverage/exceptions, and `GWDRIVE-Remediation.ps1` (runs with `-WhatIf` by default) for safe fixes to identity-map / permission gaps. Failures **inside the migration engine** (a specific file's transfer, a tool connector) are diagnosed in the tool's own docs; this guide covers the **mapping, permission, sharing, structure, and validation** issues this solution owns.

A. A migrated file has no access / lost permissions

Symptom	Cause	Remediation
File arrived but no one (or the wrong people) can open it	A source principal was **not in the identity map**, so its grant was dropped	Add the missing principal to `GWDRIVE Mapping - Identity - Source to Google`; re-apply with `GWDRIVE-Remediation.ps1` (`-WhatIf:$false`) and re-run `GWDRIVE-Validation.ps1` (Testing Test 4)
Whole scope has no grants	The identity map was incomplete for that scope's principals	Complete the map for the scope, remediate, re-validate — completeness is a gate ([CloudM — Drive permission changes](https://support.cloudm.io/hc/en-us/articles/25031144447644-Google-Drive-Permission-Changes-Microsoft-365-to-Google-Workspace))

B. Permissions were flattened (everyone is a Viewer)

- A team that should still edit lands as view-only. The **permission translation table** is wrong or incomplete for that source level. Fix `GWDRIVE Mapping - Permissions - Level to Role` so the source level maps to the correct Google role (Contribute → Contributor, manage-content → Content manager, Full Control → Manager), re-apply, and re-run Testing Test 3.

C. Team content ended up in a personal My Drive

- A SharePoint site/library was mapped to a My Drive and now has a single personal owner. Correct `GWDRIVE Mapping - Content & Structure - Source to Target` to target a **Shared Drive**, re-migrate that scope, and re-validate. Team content must never live in one person's My Drive.

D. Sharing was widened or lost

Symptom	Cause	Remediation
A file is now shared "anyone with the link" when it should not be	Anyone-link share translated without a policy check	Fix `GWDRIVE Mapping - Sharing - Link & Grant Rules` to down-scope anyone-links to domain/named and flag for owner review; re-apply and re-run Testing Test 5
A previously-shared file is no longer shared	Organization-link or grant not translated	Confirm the sharing rule and identity map cover it; re-apply and re-validate

E. Folders appear greyed out to users

• Users report visible-but-unopenable folders. This is expected Google Drive behavior for content they could not see in SharePoint — **visibility changed, not access**. Brief users to expect it; where it causes confusion, place restricted content in separate top-level folders before migrating (Best Practices).

F. A Shared Drive hit a limit

• A migration into one Shared Drive stalls or is rejected near **500,000 items**, **600 members**, or deep nesting. Split the source across multiple Shared Drives in `GWDRIVE Mapping - Content & Structure - Source to Target` (one per library / access boundary) and re-run ([Google — shared drive limits](https://support.google.com/a/answer/7338880)).

G. File counts don't reconcile

• Target item count is lower than source. Some items are legitimate tool exclusions (system files, unsupported types); others are real misses. Run `GWDRIVE-Report.ps1` for the exception list, confirm the excluded types against the tool's docs, and re-migrate the genuine misses before passing the wave (Testing Test 2).

H. The same problem recurs every wave

• A defect fixed only for one item/scope reappears. Fix the **mapping artifact** (content, permission, identity, or sharing) — not just the item — so later waves inherit the fix (Implementation Step 5). Recurring issues are almost always an un-fixed mapping, not bad luck.

I. Graph scripts fail to connect

• `GWDRIVE-Discovery.ps1`, `GWDRIVE-Validation.ps1`, `GWDRIVE-Report.ps1`, and `GWDRIVE-Set-ScopeTargeting.ps1` use Microsoft Graph against the source Microsoft 365 / Entra tenant. If connection fails, confirm the Microsoft Graph PowerShell SDK is installed and the signed-in account holds the read scopes the script requests (`Sites.Read.All`, `Files.Read.All`, `User.Read.All`, `Group.Read.All`; `Group.ReadWrite.All` for scope targeting) — consent on first

run, then re-run.

J. Diagnostics quick reference

```
.\GWDRIVE-Discovery.ps1                                        # source content/permission/sharing invento
.\GWDRIVE-Set-ScopeTargeting.ps1 -GroupName "..." -ScopeCsv .\wave.csv   # -WhatIf by default
.\GWDRIVE-Validation.ps1 -ScopeCsv .\wave.csv                  # per-item files/permissions/sharing
.\GWDRIVE-Report.ps1 -ScopeCsv .\wave.csv                      # coverage + exception list
.\GWDRIVE-Remediation.ps1 -IdentityMapCsv .\identity-map.csv   # -WhatIf by default; add -WhatIf:$false to
```

K. When to escalate

If the identity map is complete, the permission/sharing translations are correct, the tool reports the move finished, and `GWDRIVE-Validation.ps1` still reports gaps after remediation, capture the `GWDRIVE-Validation.ps1` and `GWDRIVE-Report.ps1` output and review the affected scope with the content owners, the identity/provisioning owner, and the migration-tool vendor before passing the wave.

7. Best Practices — SharePoint Online & OneDrive → Google Drive / Shared Drives Migration

Field-proven practices for moving SPO/OneDrive to Google Drive/Shared Drives **without losing access**, with a focus on shared-drive structure.

Shared-drive structure

- **Team content goes to a Shared Drive, never a personal My Drive.** A SharePoint site/library is

team-owned; putting it in one person's My Drive gives that person a kill switch over everyone's access and loses the content when they leave. Map site/library content to **Shared Drives** and personal OneDrive content to the same user's **My Drive**.

- **One Shared Drive per library (or per clean access boundary), not one giant drive.** A Shared Drive

caps at **500,000 items** and **600 members** (100 of them groups), and Google recommends staying well under **100 levels** of folder nesting ([Google — shared drive limits](https://support.google.com/a/answer/7338880)). Split large sites across drives so no drive nears a limit and each has a coherent membership.

- **Set access at the Shared Drive with groups, then vary per-folder/-file only where needed.** Add a

group as a member at the right role and let it cascade; reserve per-item permissions for the exceptions. This keeps membership legible and avoids thousands of one-off grants.

- **Put restricted content in its own top-level folder before migrating.** Google shows

previously-hidden folders **greyed out** (visible, not accessible) after migration; isolating restricted items in separate top-level folders keeps that behavior clean and predictable.

- **Keep folder trees shallow.** Deep SharePoint nesting is technically supported but hurts usability

and can complicate permission mapping — flatten obviously-over-nested trees during the mapping step.

Permissions & sharing

- **Complete the identity map before you migrate — it is the whole game.** Every source principal that holds a permission or share must resolve to a Google identity; **anything missing is silently dropped** and the file arrives with lost access ([CloudM — Drive permission changes](https://support.cloudm.io/hc/en-us/articles/25031144447644-Google-Drive-Permission-Changes-Microsoft-365-to-Google-Workspace)). Treat map completeness as a gate.
- **Map permission levels to roles deliberately — do not flatten.** Translate each SharePoint level to the closest Google role (Full Control → Manager, manage-content → Content manager, Contribute → Contributor, read+comment → Commenter, read → Viewer) against the source tenant's *actual* levels, and have owners confirm the intent is preserved ([Google — shared drive roles](https://developers.google.com/workspace/drive/api/guides/ref-roles)).
- **Apply folder-level permissions to folders, file-level to files.** This is the accurate default; reserve root/drive-level grants for cases where the top-level access should genuinely cascade to everything — over-using them over-shares.
- **Never let a migration silently widen access.** Translate anyone-with-the-link shares only where policy allows; otherwise down-scope to domain/named access and flag for owner review. A migration that broadens sharing is a security incident, not a success.
- **Tell users about the greyed-out folders.** Brief them ahead of cutover so a visible-but-locked folder reads as expected behavior, not a bug.

Method

- **Map once, migrate many.** Freeze the content, permission, identity, and sharing maps before Wave 0; deciding mapping per wave (or letting the tool default it) is the biggest source of inconsistent structure and flattened permissions.
- **Prove the whole mapping on a named pilot.** Wave 0 (a representative site + OneDrive sample) exercises files, permissions, identity, and sharing end to end — cheaper to find a mapping defect on one site than across a business unit.
- **Fix the mapping, not just the item.** Every validation exception is a defect in the content, permission, identity, or sharing artifact — fix it there so later waves inherit the fix.
- **Validate per item, every wave.** Reconcile files, check that roles match, confirm no dropped principals, and verify sharing — for every wave, before it passes. A tool's "success" is not validation.
- **One scope, one wave, fixed.** Assign every site/library/OneDrive to exactly one wave and never move a confirmed scope between waves — that risks duplicate content.
- **Cut over only after validation is clean.** Set source content read-only / decommission only once users are confirmed to have access at the mapped roles — never before.

- **Keep the tool internals with the vendor.** This solution owns the mapping, translation, and validation; the engine's console steps live in the vendor's docs — so a tool change touches the engine, not the method.

8. Deliverables Summary — SharePoint Online & OneDrive → Google Drive / Shared Drives Migration

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	The SPO/OneDrive → Drive/Shared Drives mapping model, permission/sharing translation, and the deployment model (the why)
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Google)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered steps: discover, define the mapping, pilot + per-item validate, then migrate by wave and validate
5. Testing Guide	Per-item / per-wave validation gates with pass/fail criteria
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Shared-drive structure, permissions/sharing, and method

Named mapping artifacts

Artifact	Name
Content + structure mapping	`GWDRIVE Mapping - Content & Structure - Source to Target`
Permission/role translation table	`GWDRIVE Mapping - Permissions - Level to Role`
Identity mapping table	`GWDRIVE Mapping - Identity - Source to Google`
Sharing translation rules	`GWDRIVE Mapping - Sharing - Link & Grant Rules`
Pilot scope	`GWDRIVE Pilot - Scope - Wave 0`
Per-wave migration + validation runbook	`GWDRIVE Runbook - Migrate & Validate - Per Wave`

PowerShell scripts

Script	What it does
`GWDRIVE-Discovery.ps1`	Inventories the source SharePoint sites/libraries and OneDrives: item/folder counts, structure, permission principals, and sharing (Microsoft Graph, read-only)
`GWDRIVE-Validation.ps1`	Per-item migration + permission validation: files present, roles match the translation, no dropped principals, sharing translated (read-only)
`GWDRIVE-Report.ps1`	Renders coverage and an itemized exception list: items migrated vs source and permission/sharing pass rate (read-only)
`GWDRIVE-Remediation.ps1`	Safe remediation of identity-map / permission gaps; runs with `-WhatIf` by default
`GWDRIVE-Set-ScopeTargeting.ps1`	Builds a wave's target group in Microsoft Entra from a members/scope CSV; runs with `-WhatIf` by default

The migration playbook (outline)

A single reusable playbook the migration runs from, assembled from the artifacts above:

1. **Discovery baseline** — source sites/libraries/OneDrives, counts, structure, principals, sharing.
2. **Content + structure map** — `GWDRIVE Mapping - Content & Structure - Source to Target`: which

sites/libraries become which Shared Drives; which OneDrives become which My Drives.

1. **Permission/role translation** — `GWDRIVE Mapping - Permissions - Level to Role`: each source

level → Google role, owner-confirmed.

1. **Identity map** — `GWDRIVE Mapping - Identity - Source to Google`: every principal resolved to a

Google identity (completeness is a gate).

1. **Sharing rules** — `GWDRIVE Mapping - Sharing - Link & Grant Rules`: direct grants, organization

links, and anyone-with-the-link translated deliberately.

1. **Pilot** — `GWDRIVE Pilot - Scope - Wave 0`: a representative site + OneDrive sample, validated

per item.

1. **Per-wave runbook** — `GWDRIVE Runbook - Migrate & Validate - Per Wave`: build wave group → migrate

→ validate files/permissions/sharing → report → remediate → pass.

1. **Close out / cut over** — confirm access at mapped roles, brief users on greyed-out folders, set source read-only / decommission only after validation is clean.

What the buyer receives

- This implementation guide as a Word document and PDF.
- All five PowerShell scripts in a single ZIP.
- A repeatable Drive migration following the model: Assess, then Configure, then Pilot, then Validate, then Expand.

Outcome

SharePoint Online and OneDrive content on Google Drive / Shared Drives with **permissions and sharing intact** — mapped deliberately, validated per item — proven on a pilot before scaling, and migrated wave by wave so files land in the right shared drives with the right access.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

GWDRIVE-Discovery.ps1

Purpose: Reads the in-scope SharePoint Online sites/document libraries and OneDrive for Business drives through Microsoft Graph and reports item/folder counts, the distinct permission principals (users, groups) and the sharing in place. Read-only — it supports the Assess step, moves no content, and changes no permissions. Its principal list is the input to the identity mapping table (map every principal to a Google identity before migrating, or its permission is dropped). Placeholders use plain text such as your-tenant-id, not angle brackets. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWDRIVE-Discovery.ps1 Optional: -SiteIds site-id-1,site-id-2 to scope to named sites (default: all sites). Optional: -CsvPath .\gwdrive-discovery.csv to export the per-scope summary. Requires the Microsoft Graph PowerShell SDK. On first run, consent to the read scopes: Sites.Read.All, Files.Read.All, User.Read.All, Group.Read.All. EXPECTED RESULTS A per-scope summary (site/library and OneDrive): item and folder counts, distinct permission principals, and shared-item counts, plus a PASS/GAP line per signal you record against the Requirements checklist and use to build the mapping artifacts. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string[]]$SiteIds, [string]$CsvPath)

How to run: powershell -ExecutionPolicy Bypass -File .\GWDRIVE-Discovery.ps1 Optional: -SiteIds site-id-1,site-id-2 to scope to named sites (default: all sites). Optional: -CsvPath .\gwdrive-discovery.csv to export the per-scope summary. Requires the Microsoft Graph PowerShell SDK. On first run, consent to the read scopes: Sites.Read.All, Files.Read.All, User.Read.All, Group.Read.All. EXPECTED RESULTS A per-scope summary (site/library and OneDrive): item and folder counts, distinct permission principals, and shared-item counts, plus a PASS/GAP line per signal you record against the Requirements checklist and use to build the mapping artifacts. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string[]]$SiteIds, [string]$CsvPath)

GWDRIVE-Remediation.ps1

Purpose: Closes the mapping gaps that cause dropped or flattened permissions: it takes the unmapped source principals a validation run surfaced and proposes an identity-map row for each by resolving the source user/group against Microsoft Graph and suggesting the matching Google primary address, and it appends any missing source permission levels to the role-translation table with a suggested Google role. It writes only to the local mapping CSVs — it never changes source or target permissions and never touches the migration tool. Runs with -WhatIf by default; add -WhatIf:$false to write the CSV rows. Placeholders use plain text such as your-domain.example, not angle brackets. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\GWDRIVE-Remediation.ps1 `-UnmappedCsv .\unmapped-principals.csv -IdentityMapCsv .\identity-map.csv` -TargetDomain your-domain.example Apply: ...\GWDRIVE-Remediation.ps1 -UnmappedCsv .\unmapped-principals.csv ` -IdentityMapCsv .\identity-map.csv -TargetDomain your-domain.example -WhatIf:$false UnmappedCsv column: Principal (source email or group:Name). IdentityMapCsv columns: Source,Google. Requires the Microsoft Graph PowerShell SDK; consent to User.Read.All, Group.Read.All. EXPECTED RESULTS In preview: the identity-map rows that would be appended (source principal -> suggested Google address) and any level rows to add. When applied: those rows are written to the mapping CSVs for review before the next validation run. Suggestions are proposals — confirm each before use. .NOTES White-label — the

repeatable way. Writes only local mapping CSVs; -WhatIf by default. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'Medium')] param([Parameter(Mandatory)][string]$UnmappedCsv, [Parameter(Mandatory)][string]$IdentityMapCsv, [string]$TargetDomain = 'your-domain.example', [string]$PermissionMapCsv, [string[]]$UnmappedLevels)

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\GWDRIVE-Remediation.ps1 `-UnmappedCsv .\unmapped-principals.csv -IdentityMapCsv .\identity-map.csv` -TargetDomain your-domain.example Apply: ...\GWDRIVE-Remediation.ps1 -UnmappedCsv .\unmapped-principals.csv ` -IdentityMapCsv .\identity-map.csv -TargetDomain your-domain.example -WhatIf:$false UnmappedCsv column: Principal (source email or group:Name). IdentityMapCsv columns: Source,Google. Requires the Microsoft Graph PowerShell SDK; consent to User.Read.All, Group.Read.All. EXPECTED RESULTS In preview: the identity-map rows that would be appended (source principal -> suggested Google address) and any level rows to add. When applied: those rows are written to the mapping CSVs for review before the next validation run. Suggestions are proposals — confirm each before use. .NOTES White-label — the repeatable way. Writes only local mapping CSVs; -WhatIf by default. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'Medium')] param([Parameter(Mandatory)][string]$UnmappedCsv, [Parameter(Mandatory)][string]$IdentityMapCsv, [string]$TargetDomain = 'your-domain.example', [string]$PermissionMapCsv, [string[]]$UnmappedLevels)

GWDRIVE-Report.ps1

Purpose: Renders migration coverage and an itemized exception list for the wave's scopes: source item/folder counts per scope, the count of distinct permission principals, and the sharing in place — so you can reconcile what the migration should move and see the exceptions (unmapped principals, over-broad sharing, empty scopes) before passing the wave. Read-only against Microsoft Graph; it moves nothing and changes nothing. Placeholders use plain text such as your-tenant-id, not angle brackets. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWDRIVE-Report.ps1 -ScopeCsv .\wave.csv Optional: -IdentityMapCsv .\identity-map.csv to flag principals with no Google mapping. Optional: -CsvPath .\gwdrive-coverage.csv to export the per-scope coverage table. ScopeCsv columns: Kind (SharePoint|OneDrive), Scope, DriveId. Requires the Microsoft Graph PowerShell SDK; consent to Sites.Read.All, Files.Read.All, User.Read.All, Group.Read.All. EXPECTED RESULTS A per-scope coverage table (files, folders, principals, shared items) and an exception list. The wave is ready to pass when coverage reconciles and the exception list is empty. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([Parameter(Mandatory)][string]$ScopeCsv, [string]$IdentityMapCsv, [string]$CsvPath)

How to run: powershell -ExecutionPolicy Bypass -File .\GWDRIVE-Report.ps1 -ScopeCsv .\wave.csv Optional: -IdentityMapCsv .\identity-map.csv to flag principals with no Google mapping. Optional: -CsvPath .\gwdrive-coverage.csv to export the per-scope coverage table. ScopeCsv columns: Kind (SharePoint|OneDrive), Scope, DriveId. Requires the Microsoft Graph PowerShell SDK; consent to Sites.Read.All, Files.Read.All, User.Read.All, Group.Read.All. EXPECTED RESULTS A per-scope coverage table (files, folders, principals, shared items) and an exception list. The wave is ready to pass when coverage reconciles and the exception list is empty. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([Parameter(Mandatory)][string]$ScopeCsv, [string]$IdentityMapCsv, [string]$CsvPath)

GWDRIVE-Set-ScopeTargeting.ps1

Purpose: Creates (or reuses) the Microsoft Entra security group that defines a migration wave — the pilot (Wave 0) first, then each expansion wave — so every wave runs against a bounded, well-understood population of OneDrive owners and site members, and validation/reporting can key off it. Optionally seeds members from a CSV of user UPNs (the OneDrive owners and site members in the wave's scope). This is the only included script that writes to the tenant: it creates a group and adds members, nothing else — no content moved, no permissions changed. Runs with -WhatIf by default. Placeholders use plain text such as your-domain.example, not angle brackets. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\GWDRIVE-Set-ScopeTargeting.ps1 `-GroupName "GWDRIVE Wave - Wave 0 - Pilot" -MembersCsv .\wave0-users.csv Apply: ...\GWDRIVE-Set-ScopeTargeting.ps1` -GroupName "GWDRIVE Wave - Wave 0 - Pilot" -MembersCsv .\wave0-users.csv ` -WhatIf:$false MembersCsv is a CSV with an 'Upn' column. Requires the Microsoft Graph PowerShell SDK and Group.ReadWrite.All (and User.Read.All to resolve UPNs). EXPECTED RESULTS In preview: the group that would be created/reused and the members that would be added. When applied: the security group exists and contains the listed members, ready to target with the wave's migrate-and-validate runbook. .NOTES White-label — the repeatable way. Writes to Microsoft Entra; -WhatIf by default. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([Parameter(Mandatory)][string]$GroupName, [string]$MembersCsv)

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\GWDRIVE-Set-ScopeTargeting.ps1 `-GroupName "GWDRIVE Wave - Wave 0 - Pilot" -MembersCsv .\wave0-users.csv Apply: ...\GWDRIVE-Set-ScopeTargeting.ps1` -GroupName "GWDRIVE Wave - Wave 0 - Pilot" -MembersCsv .\wave0-users.csv ` -WhatIf:$false MembersCsv is a CSV with an 'Upn' column. Requires the Microsoft Graph PowerShell SDK and Group.ReadWrite.All (and User.Read.All to resolve UPNs). EXPECTED RESULTS In preview: the group that would be created/reused and the members that would be added. When applied: the security group exists and contains the listed members, ready to target with the wave's migrate-and-validate runbook. .NOTES White-label — the repeatable way. Writes to Microsoft Entra; -WhatIf by default. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([Parameter(Mandatory)][string]$GroupName, [string]$MembersCsv)

GWDRIVE-Validation.ps1

Purpose: Validates a wave per item against the mapping: confirms the source scopes were fully inventoried (files/folders and their permission principals), and checks each source principal resolves in the identity map so no permission will be dropped, and each source permission level resolves in the role translation table so no role will be flattened. Read-only against Microsoft Graph on the source side; it reads the identity-map and permission-map CSVs to confirm coverage. It changes nothing. Placeholders use plain text such as your-tenant-id, not angle brackets. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWDRIVE-Validation.ps1 -ScopeCsv .\wave.csv ` -IdentityMapCsv .\identity-map.csv -PermissionMapCsv .\permission-map.csv ScopeCsv columns: Kind (SharePoint|OneDrive), DriveId. IdentityMapCsv column: Source (source principal email or group:Name). PermissionMapCsv column: SourceLevel. Requires the Microsoft Graph PowerShell SDK; consent to Sites.Read.All, Files.Read.All, User.Read.All, Group.Read.All. EXPECTED RESULTS A per-scope line and a wave summary: items readable, principals covered by the identity map vs unmapped (would be dropped), and permission levels covered by the translation vs unmapped (would be flattened). PASS when every principal and level is covered; GAP lists what to fix. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([Parameter(Mandatory)][string]$ScopeCsv, [string]$IdentityMapCsv, [string]$PermissionMapCsv)

How to run: powershell -ExecutionPolicy Bypass -File .\GWDRIVE-Validation.ps1 -ScopeCsv .\wave.csv ` -IdentityMapCsv .\identity-map.csv -PermissionMapCsv .\permission-map.csv ScopeCsv columns: Kind (SharePoint|OneDrive), DriveId. IdentityMapCsv column: Source (source principal email or group:Name). PermissionMapCsv column: SourceLevel. Requires the Microsoft Graph PowerShell SDK; consent to Sites.Read.All, Files.Read.All, User.Read.All, Group.Read.All. EXPECTED RESULTS A per-scope line and a wave summary: items readable, principals covered by the identity map vs unmapped (would be dropped), and permission levels covered by the translation vs unmapped (would be flattened). PASS when every principal and level is covered; GAP lists what to fix. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([Parameter(Mandatory)][string]$ScopeCsv, [string]$IdentityMapCsv, [string]$PermissionMapCsv)